



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Advanced Level

MATHEMATICS
PAPER 2

6082/2
3 hours

SPECIMEN PAPER

Additional materials:
Answer booklet
Graph paper
List of Formulae MF 7
Non programmable Scientific calculator
Mathematical instruments

INSTRUCTIONS TO CANDIDATES

Write your Name, Centre number and Candidate number in the spaces provided on the answer booklet.

Answer **all** questions in **Section A** and any **five** from **Section B** and **C**.
All working must be clearly shown.

If a numerical answer cannot be given exactly, and the accuracy required is not specified in the question, then in the case of an angle it should be given to the nearest degree, and in other cases it should be given correct to 2 significant figures.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [] at the end of each question or part question.
The total number of marks for this paper is 120.

This specimen paper consists of 10 printed pages.
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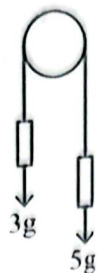
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SECTION A [40 Marks]

Answer **all** questions in this section.

- 1 Two bags of sand of masses 5 kg and 3 kg are connected by a light inelastic string over a smooth fixed pulley. The system moves freely.



Find the

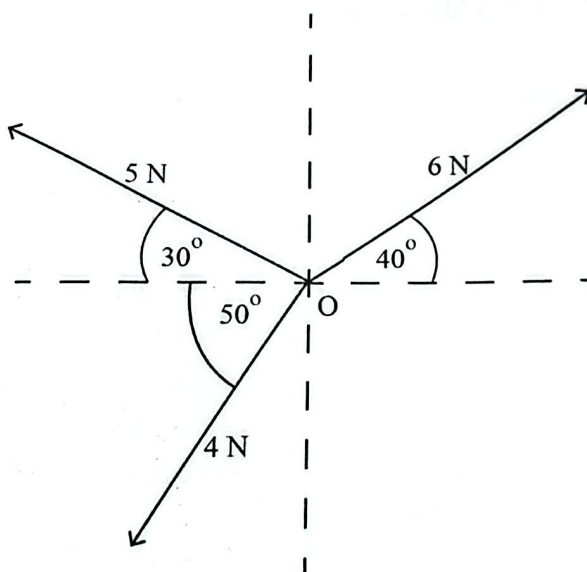
(a) acceleration of the bags,

[2]

(b) tension in the string.

[2]

- 2 The diagram shows a system of coplanar forces P, Q and S of magnitudes 5 N, 4 N and 6 N, respectively, acting on a particle at point O.



Find the magnitude and direction of the resultant force.

[5]



- 3 A particle of mass 5 kg rests on a rough plane inclined at 30° to the horizontal and the coefficient of friction between the particle and the plane is $\frac{\sqrt{3}}{10}$.

Find the force P, acting parallel to the plane, which must be applied to the particle in order to just prevent the motion down the plane. [5]

- 4 An equilateral triangle ABC of side 3 m has forces of magnitudes P, Q and R acting along sides AB, BC and CA, respectively. The anticlockwise moments about the axes perpendicular to the triangle through A, B and C are $+6\sqrt{3}$ Nm, $-12\sqrt{3}$ Nm and $+3\sqrt{3}$ Nm, respectively.

Calculate the magnitudes P, Q and R. [6]

- 5 State

(a) **one** advantage and **one** disadvantage of using the median as a measure of central tendency. [2]

(b) **one** advantage and **one** disadvantage of a stem and leaf diagram in data presentation. [2]

- 6 (a) State any **two** measures of dispersion. [2]

(b) The shoe sizes of Tongogara High soccer team are as follows:

5; 7; 6; 5; 4; 8; 9; 7; 6; 8; 9

Calculate the

(i) range [1]

(ii) standard deviation. [2]

- 7 In a group of 80 village elders, 20 own cattle, 35 own sheep and 10 own both cattle and sheep. One elder is chosen at random from the group for the position of a committee member in the Pfumvudza Committee.

Find the probability that the person owns

(a) neither sheep nor cattle, [1]

(b) either sheep or cattle, [2]

(c) cattle given that he owns sheep. [2]



- 8 X is a random variable such that the probability of tossing a biased coin to get a head is $\frac{2}{3}$.

Given that $X_n \sim \text{Geo}(p)$, find

- (a) $P(X \leq 2)$, [3]
- (b) $P(X > 7/X > 3)$. [3]

SECTION B (Mechanics)

Do **not** answer more than **three** questions in this section.

- 9 A footballer at Kete High school kicks a ball from a point A on level ground with an initial speed of 45 m/s at an angle θ above the horizontal. The ball strikes the ground at point B which is 90 m from A on the same horizontal level as A.

Modeling the ball as a particle and neglecting air resistance, find the

- (a) two possible angles of projection, [5]
 - (b) difference between the two times of flight generated by the two possible angles of projection in part (b), [3]
 - (c) magnitude and direction of the speed of the ball when $t = 1.5$ seconds, [6]
 - (d) maximum height reached by the ball using the smaller angle of projection. [2]
- 10 (a) Two forces F_1 and F_2 are represented by $(2i + 3j)$ N and $(-6i - 6j)$ N, respectively. Find the modulus and direction of $F_1 + F_2$. [4]
- (b) Two forces have magnitudes A and B and the angle between them is θ where θ is an acute angle. The resultant of these forces has magnitude R, and makes an angle α with force A,

Show that

(i) $R^2 = A^2 + B^2 + 2AB\cos\theta$, [3]

(ii) $\tan \alpha = \frac{B\sin\theta}{A+B\cos\theta}$. [7]

(iii) If $A = B$ and $\theta = 30^\circ$, find the value of α in part (b) (ii). [2]

- 11 A car moves in a straight line such that

$$v = 3t^2 - 14t + 8.$$

Initially the displacement of the car from a fixed point O on the line is 40 m.

Find the

- (a) initial velocity of the car, [2]
- (b) values of t when the car is at rest, [4]
- (c) acceleration of the car when $t = 3$ s, [4]



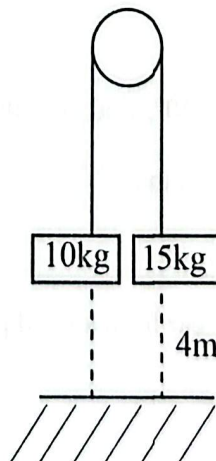
- (d) displacement of the car from point O when $t = 2$ s. [6]

12 A particle is initially at rest at point A on a straight line ABCD. The particle moves from A to B with uniform acceleration, reaching B with a speed of 14 m/s after 2 seconds. It accelerates uniformly to 3 m/s^2 and reaches C, 2 seconds after passing B. The particle then retards uniformly to come to rest at D after a further 8 seconds.

- (a) Sketch the velocity-time graph for the motion [4]
- (b) Find the
- (i) acceleration of the particle from A to B, [2]
 - (ii) speed of the particle at C, [2]
 - (iii) retardation of the particle from C to D, [2]
 - (iv) speed of the particle when $t = 10$ seconds, [2]
 - (v) total distance from A to D, [2]
 - (vi) average speed. [2]

13 (a) State Newton's 3 laws of motion. [3]

- (b) Packets of maize meal of masses 10 kg and 15 kg are connected by a light inextensible string and hang vertically on either side of the smooth pulley. The system is released from rest with both packets at a height of 4 m above ground level.



Find the

- (i) acceleration of the packets in terms of g , [3]
- (ii) tension in the string, [2]

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- (iii) force exerted on the pulley, [2]
- (iv) total distance travelled by the 10 kg packet upwards. [6]



SECTION C (Statistics)

Do not answer more than three questions in this section.

- 14 A fair coin and an unbiased six-sided die are thrown simultaneously. Let x be a random variable defined as follows:

If the coin shows a head, then X is the score on the die. If the coin shows a tail, then X is twice the score on the die.

- (a) (i) Draw the outcome table. [4]
 (ii) Find $E(X)$. [4]
 (b) Show that $\text{Var}(X) = \frac{497}{48}$. [4]

- (c) The experiment is repeated and the sum of the two values obtained for X are denoted by Y .

Find $P(Y = 4)$. [4]

- 15 (a) The heights of 100 Moringa plants are measured to the nearest centimetre. The plants are classified according to the year they were planted. The table summarises the data collected.

Year planted	2025	2024	2023	2022	2021	202
Height (cm)	0-50	51-100	101-150	151-200	201-300	301-400
Frequency	3	11	22	44	16	4

- (i) Draw a cumulative frequency curve for the information, using a scale of 2 cm to represent 50 cm on the horizontal axis and 2 cm to represents 10 Moringa plants on the vertical axis. [4]
 (ii) Using the graph in (a)(i), draw a box and whisker plot. [4]
 (b) The random variable Y represents ages of some Solusi University students. X is defined as the remainder, when each age is divided by 3.

$$Y = \{20; 21; 22; \dots 29\}$$

- (i) Write down the values of X . [1]
 (ii) Draw a probability distribution table for X . [2]
 (iii) Find $E(X)$ and $\text{Var}(X)$. [5]



- 16 (a) Distinguish between a *continuous random variable* and a *discrete random variable*. [2]

- (b) Let X be a continuous random variable with probability density function defined by

$$f(x) = \begin{cases} c(4+x) & 0 \leq x \leq 4 \\ 0 & \text{otherwise} \end{cases}$$

- (i) Show that $c = \frac{3}{160}$. [3]
- (ii) Find $E(X)$ and $P(X \geq 2)$. [5]
- (iii) Show that $P(1 \leq X \leq 3) = \frac{37}{80}$. [2]
- (c) The height (X) of 100 army officers recruits in Zimbabwe were discovered to follow a normal distribution with mean and standard deviation 180 cm and 3 cm, respectively.

Find $P(175 < X < 178)$. [4]

- 17 (a) State the mean and variance of a random variable X that follows a geometric distribution with parameter P . [2]
- (b) State the condition for the application of a geometric distribution [2]
- (c) At a school the probability that a candidate passes Mathematics is $\frac{3}{5}$. A candidate continues to write Mathematics examination until a pass is obtained.
- (i) Find the expected number of attempts needed to pass and variance of the number of attempts to pass. [3]
- (ii) Calculate the probability that a candidate needs at least 4 attempts to pass. [2]
- (iii) Find the probability that a pass is achieved between 3 and 5 attempts inclusively. [2]
- (iv) What is the most probable number of attempts required to pass. [1]
- (d) Given that $X \sim \text{Geo}(p)$ and $E(X) = 5$, find p and the variance of X . [4]



- 18 (a) The number of accidents occurring in a week along Masvingo-Beitbridge road follow a Poisson distribution with variance 5.

Find the probability that

- (i) Exactly 3 accidents occur per week. [3]
 - (ii) Exactly 10 accidents occur in 2 weeks. [3]
 - (iii) At least 3 accidents occur in 2 weeks to 3 decimal places. [3]
- (b) Daramombe High school has intensified its cattle ranching project. It is believed that $\frac{6}{10}$ of the pregnant cows give birth normally.

In a sample of 50 pregnant cows, and using a suitable approximation, find the probability that

- (i) 25 are normal births, [4]
- (ii) normal births are between 30 and 35 inclusively. [3]

